When iterated $n+1$ times, we can see the scheme $[\![\text{OABA}]\!]$ produces a mapping of the form

$$\left(\mathscr{U}_h^{[\![\text{OABA}]\!]}\right)^{n+1} = \mathscr{U}_{h/2}^{\text{A}} \circ \mathscr{U}_{h/2}^{\text{B}} \circ \left(\mathscr{U}_h^{[\![\text{BAOAB}]\!]}\right)^{n} \circ \mathscr{U}_{h/2}^{\text{B}} \circ \mathscr{U}_{h/2}^{\text{A}} \circ \mathscr{U}_h^{\text{O}},$$

therefore through a coordinate transformation the $[\![\text{OABA}]\!]$ method can be morphed into the $[\![\text{BAOAB}]\!]$ scheme. To put this another way, if we compute an iteration using $[\![\text{OABA}]\!]$ and simply evaluate it at different points within the step, we can recover the same results as the $[\![\text{BAOAB}]\!]$ method would provide. Another term to describe this relationship is to say that one method is *conjugate* to the other.

7.3.2 Other Langevin Integrators

We may use our alphabet of A, B and O splitting pieces to create a wide variety of methods other than those given in the previous section (some requiring multiple force evaluations). However, it is not obvious that there is any gain in increasing the complexity of the splitting scheme (in terms of the length of its codifying string). Some popular methods can be rewritten in terms of this splitting, for example the scheme considered in [62] is equivalent to $[\![\text{OBABO}]\!]$. Examples of methods based on this type of splitting were also proposed by De Fabritiis et al [96] and Thalmann and Farago [363], in the context of dissipative particle dynamics (see Chapter 8).

Other Langevin dynamics splitting schemes in common use utilize a different additive decomposition of the SDE vector field. For example, the Stochastic Position Verlet (SPV) method of [265] relies on the splitting strategy

$$\mathrm{d}\begin{bmatrix}\boldsymbol{q}\\ \boldsymbol{p}\end{bmatrix} = \begin{bmatrix}\boldsymbol{M}^{-1}\boldsymbol{p}\\ \mathbf{0}\end{bmatrix}\mathrm{d}t + \begin{bmatrix}\mathbf{0}\\ -\nabla U(\boldsymbol{q})\mathrm{d}t - \gamma\boldsymbol{p}\mathrm{d}t + \sigma\boldsymbol{M}^{1/2}\mathrm{d}\mathbf{W}\end{bmatrix}.$$

To solve the splitting piece including the stochastic $\mathrm{d}\mathbf{W}$ term, we note that $\boldsymbol{q}$ does not change through its integration, and hence we may treat $\boldsymbol{F} = -\nabla U$ as constant during this part of the step. The exact solution is hence given through the usual Ornstein-Uhlenbeck formula, where the scheme itself derives from a symmetric integration of the two parts:

Stochastic Position Verlet (SPV)

$$\begin{aligned}\boldsymbol{q}_{n+1/2} &= \boldsymbol{q}_n + (h/2)\boldsymbol{M}^{-1}\boldsymbol{p}_n\\ \boldsymbol{p}_{n+1} &= e^{-h\gamma}\boldsymbol{p}_n - \eta\nabla U(\boldsymbol{q}_n) + \zeta\boldsymbol{M}^{1/2}\mathbf{R}_n\\ \boldsymbol{q}_{n+1} &= \boldsymbol{q}_{n+1/2} + (h/2)\boldsymbol{M}^{-1}\boldsymbol{p}_{n+1}\end{aligned}$$

where $\eta = (1-e^{-h\gamma})/\gamma$, $\zeta = \left[k_{\text{B}}\text{T}\left(1-e^{-2\gamma h}\right)\right]^{1/2}$. It is important to note that in the limit of infinite friction, we have $\eta \to 0$ independent of the timestep h. This suggests that this splitting strategy is unsuitable for large choices of the parameter γ as the